

Code Quality & Reusability Assessment

|              |              |               |                   |
|--------------|--------------|---------------|-------------------|
| Date         | 06 July 2026 | Team ID       | PNT2026TMID124356 |
| Project Name | Smart Lender | Maximum Marks | 5 Marks           |

| Code Layout Checklist |                        |                                          |                   |                                                    |
|-----------------------|------------------------|------------------------------------------|-------------------|----------------------------------------------------|
| S.No                  | Code Quality Parameter | Description                              | Followed (Yes/No) | Remarks                                            |
| 1                     | Consistent Indentation | Uniform spacing/tabs used throughout     | Yes               | 4 spaces standard in Python files                  |
| 2                     | Proper File Structure  | Logical organization of source files     | Yes               | Clear split of train, app, static, templates       |
| 3                     | Meaningful Variables   | Reflects variable purpose clearly        | Yes               | e.g. coapplicant_income_log, features_scaled       |
| 4                     | Method Names           | Functions are descriptively named        | Yes               | e.g. preprocess_input(), load_model()              |
| 5                     | Code Comments          | Inline comments explaining complex logic | Yes               | Detailed math comments on risk flags and SMOTE     |
| 6                     | Modular Design         | Split into reusable functions/modules    | Yes               | Preprocessing isolated in helper functions         |
| 7                     | No Redundant Code      | Duplicate or dead code removed           | Yes               | Clean code structures, unused model trials removed |
| 8                     | Error Handling         | Exceptions handled gracefully            | Yes               | Checks for missing rdf.pkl, invalid CSV schema     |

| Reusable Components Summary |
|-----------------------------|
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| S.No | Component Name       | Language/Tech | Where Reused                                     | Reusability Level |
|------|----------------------|---------------|--------------------------------------------------|-------------------|
| 1    | preprocess_input()   | Python        | Both app.py and train.py for preprocessing       | High              |
| 2    | StandardScaler asset | scikit-learn  | Fit in training, transformed during live scoring | High              |
| 3    | XGBoost classifier   | XGBoost       | Serialized and loaded on flask startup           | High              |